

FIT3143 Applied 2

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Image Rotation with GPU

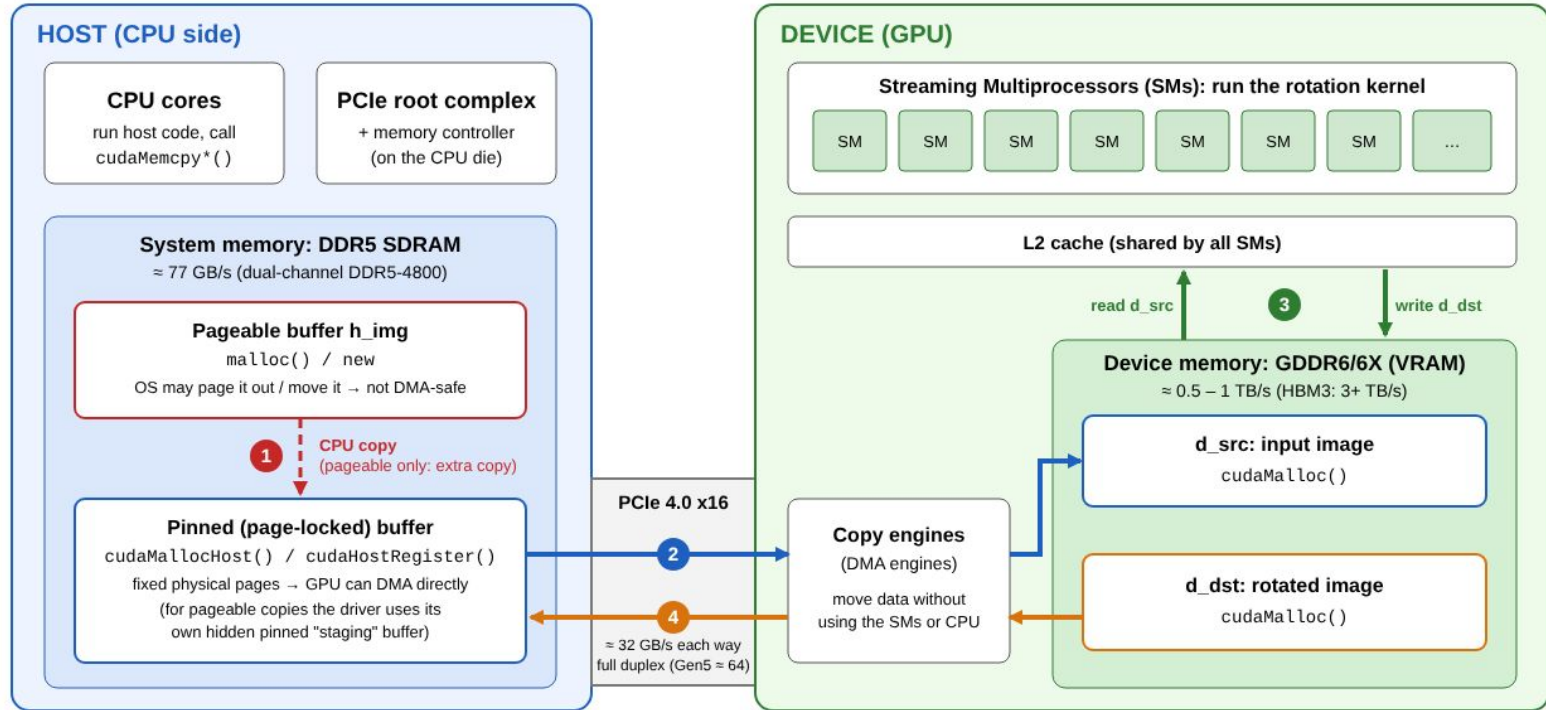
Moving image data from the CPU to the GPU

A discrete GPU has its own memory (GDDR), separate from the system's DDR5. The image must be copied across before the GPU can work on it, and the result copied back.

GPUs use GDDR for its much higher bandwidth (~ 1 TB/s vs ~ 77 GB/s), which feeds thousands of threads reading at once.

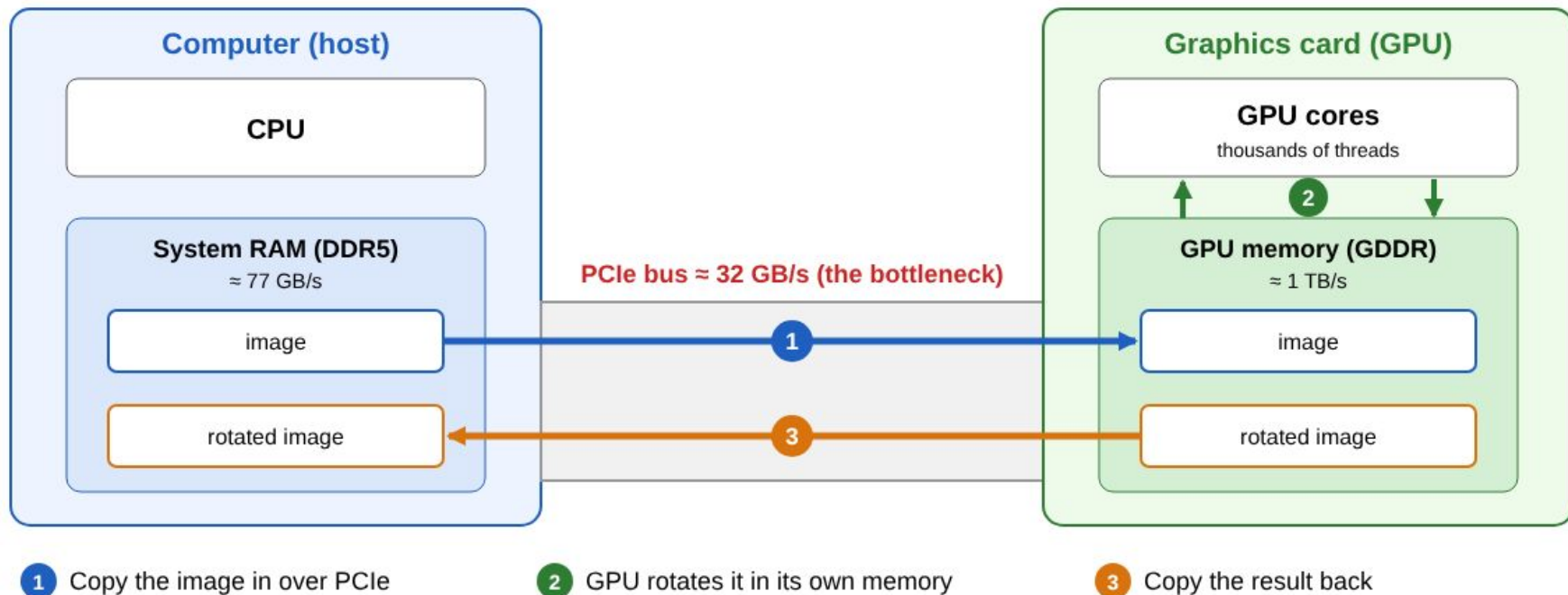
The GPU's memory system trades higher latency for that bandwidth, and hides the latency by switching between thousands of threads.

Moving an image between host memory (DDR5) and GPU memory (GDDR)



- 1 Pageable only:** the CPU copies `h_img` into a pinned staging buffer (extra copy that uses CPU time and memory bandwidth).
- 2 Host $\rightarrow$ Device:** a GPU copy engine DMA's the pinned pages over PCIe into `d_src` (`cudaMemcpyHostToDevice`).
- 3 On device:** SMs read `d_src` and write `d_dst` through L2 at VRAM speed ($0.5-1 \text{ TB/s}$): roughly $15-30\times$ faster than the PCIe link.
- 4 Device $\rightarrow$ Host:** a copy engine DMA's `d_dst` back over PCIe (`cudaMemcpyDeviceToHost`); pageable targets get another CPU copy.

Moving the image to the GPU and back



PCIe is $\sim 30\times$ slower than GPU memory, so the copies take longer than the rotation.

Why not just...

- ...have the GPU read it from system memory?
 - The GPU would have to read it at 32 GB/s (PCIe speed) instead of 1 TB/s, about 30x slower.
 - Each read takes much longer, as it has to make a round trip across the bus.
 - It's better to make one big copy, rather than many smaller ones.
- ...keep everything in GPU memory?
 - You cannot start at the GPU, as the OS, the program and files are loaded into system memory.
 - Results would have to be sent back to the CPU anyway.
- ...use one memory for both?
 - My (Nam) laptop does that! I have a laptop with an M5 chip with 24 GB of unified memory.
 - The tradeoff is a lower peak bandwidth than a top discrete GPU (RTX 5090, PRO 6000, B200 etc), the memory bandwidth of an M5 chip is 153.6 GB/s compared to 1TB/s+ discrete GPUs can reach.
 - CPU and GPU would compete for the same memory and bandwidth.